
Integration Manual

for MPC560XB MCU Driver

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Rev 1.2





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Chapter 1

Revision History

Table 1-1. Revision History

Revision	Date	Author	Description
1.0	17/02/2012	Francesco Mazzearella	Initial Release
1.1	13/05/2013	Swarupa Chaudhury	RTM 1.0.0 Release
1.2	28/05/2014	Ronan Forest	RTM 1.0.1 Release



Chapter 2

Introduction

This integration manual describes the integration requirements for MCU Driver for MPC560XB microcontrollers.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of Freescale Semiconductor .

Table 2-1. MPC560XB Derivatives

Freescale Semiconductor	mpc5601dxlh_lqfp64 mpc5601dxll_lqfp100 mpc5602dxlh_lqfp64 mpc5602dxll_lqfp100 mpc5602bxlh_lqfp64 mpc5602bxll_lqfp100 mpc5602bxlq_lqfp144 mpc5602cxlh_lqfp64 mpc5602cxll_lqfp100 mpc5603bxlh_lqfp64 mpc5603bxll_lqfp100 mpc5603bxlq_lqfp144 mpc5603cxlh_lqfp64 mpc5603cxll_lqfp100 mpc5604bxlh_lqfp64 mpc5604bxll_lqfp100 mpc5604bxlq_lqfp144 mpc5604bxmg_mapbga208 mpc5604cxlh_lqfp64 mpc5604cxll_lqfp100 mpc5605bxll_lqfp100 mpc5605bxlq_lqfp144 mpc5605bxlu_lqfp176 mpc5606bkll_lqfp100 mpc5606bxlq_lqfp144 mpc5606bxlu_lqfp176 mpc5607bxlu_lqfp176 mpc5607bxmg_mapbga208
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All of the above microcontroller devices are collectively named as MPC560XB .

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About this Manual

This Technical Reference employs the following typographical conventions:

Boldface type: Bold is used for important terms, notes and warnings.

Italic font: Italic typeface is used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

2.4 Acronyms and Definitions

Table 2-2. Acronyms and Definitions

Term	Definition
API	Application Programming Interface
AUTOSAR	Automotive Open System Architecture
ASM	Assembler
BSMI	Basic Software Make file Interface
BSW	Basic Software
CAN	Controller Area Network
DEM	Diagnostic Event Manager
DET	Development Error Tracer
DMA	Direct Memory Access
ECU	Electronic Control Unit
EcuM	ECU state Manager
C/CPP	C and C++ Source Code
GUI	Graphical User Interface
ISR	Interrupt Service Routine
OS	Operating System
PB Variant	Post Build Variant
PC Variant	Pre Compile Variant
VLE	Variable Length Encoding
N/A	Not Applicable
MCU	Micro Controller Unit

2.5 Reference List

Table 2-3. Reference List

#	Title	Version
1	AUTOSAR 4.0 Rev0003MCU Driver Software Specification Document.	3.2.0 Rev 3
2	MPC5607B Reference Manual	Rev. 7.2, May 2012
3	MPC5604B Reference Manual	Rev. 8.2, September 2013
4	5606B Reference Manual	Rev. 2, May 2014
5	5602D Reference Manual	Rev. 4.2, Dec 2013

Chapter 3

Building the Driver

This section describes the source files and various compilers, linker options used for building the Autosar MCU driver for Freescale Semiconductor MPC560XB . It also explains the EB Tresos Studio plugin setup procedure.

3.1 Build Options

The MCU driver files are compiled using

- Green Hills MULTI v6.1.4
- Windriver DIAB Rel DIAB_5_9_2_0-FCS_20120725_234511
- Freescale CodeWarrior for MPC5xxx v2.10 - compiler Version 4.3 build 224
- HighTec v4.6.4.0, Compiler gcc v4.6.3

The compiler, linker flags used for building the driver are explained below:

Note

The TS_T2D13M10I1R0 plugin name is composed as follow:

TS_T = Target_Id

D = Derivative_Id

M = SW_Version_Major

I = SW_Version_Minor

R = Revision

(i.e. Target_Id = 2 identifies PowerPC architecture and
Derivative_Id = 13 identifies the MPC560XB)

3.1.1 CW Compiler/Linker/Assembler Options

Table 3-1. Compiler Options

Option	Description
-proc Zen	Generates and links object code for Zen processor. The compiler uses unsigned as the default parameter for the -char switch
-lang c	Expects source code to conform to the language specified by the ISO/IEC 9899-1990 ("C90") standard
-opt all	This option is selected all optimization (the same as -opt speed,level=4,intrinsics,noframe)
-common off	Disables moving uninitialized data into a common section
-sdatathreshold 0	Specifies the threshold size (in bytes) for an item considered by the linker to be small data. (The linker stores small data items in the Small Data address space. The compiler can generate faster code to access this data.)
-sdata2threshold 0	Specifies the threshold size (in bytes) for an item considered by the linker to be small constant data. (The linker stores small constant data items in the Small Constant Data address space.)
-vle	Tells the compiler and linker to generate and lay out Variable Length Encoded (VLE) instructions, available on Zen variants of Power Architecture processors
-use_lmw_stmw on	Enables the use of multiple load and store instructions for function prologues and epilogues
-ppc_asm_to_vle	Converts regular Power Architecture assembler mnemonics to equivalent VLE (Variable Length Encoded) assembler mnemonics in the inline assembler
-Cpp_exceptions off	When on, generates executable code for C++ exceptions. When off, generates smaller, faster executable code
-func_align 4	Specifies alignment of functions in executable code
-sym dwarf-2,full	Generate DWARF-2-conforming debugging information (Debug With Arbitrary Record Format)
-gdwarf-2	Generate DWARF-2-conforming debugging information (Debug With Arbitrary Record Format). The linker ignores debugging information that is not in the Dwarf 1, Dwarf 2 format
-w on	Turns on most warning messages
-r	Compiler should expect function prototypes
-w undefmacro	Issues warning messages on the use of undefined macros in #if and #elif conditionals
-char unsigned	Controls the default sign of the char data type: char data items are unsigned
-nosyspath	Performs a search of both the user and system paths, treating #include statements of the form #include xyz the same as the form #include "xyz"
-fp none	No floating point code generation
-DAUTOSAR_OS_NOT_USED	-D defines a preprocessor symbol and optionally can set it to a value. AUTOSAR_OS_NOT_USED: By default in the package, the drivers are compiled to be used without Autosar OS. If the drivers are used with Autosar OS, the compiler option '-DAUTOSAR_OS_NOT_USED' must be removed from project options
-DMWERKS	-D defines a preprocessor symbol and optionally can set it to a value. This one defines the CW preprocessor symbol.

Table 3-2. Assembler Options

Option	Description
-proc Zen	Generates and links object code for Zen processor. The compiler uses unsigned as the default parameter for the -char switch

Table continues on the next page...

Table 3-2. Assembler Options (continued)

Option	Description
-vle	Tells the compiler and linker to generate and lay out Variable Length Encoded (VLE) instructions, available on Zen variants of Power Architecture processors
-sym dwarf-2,full	Generate DWARF-2-conforming debugging information (Debug With Arbitrary Record Format)
-gdwarf-2	Generate DWARF-2-conforming debugging information (Debug With Arbitrary Record Format). The linker ignores debugging information that is not in the Dwarf 1, Dwarf 2 format.
-DMWERKS	-D defines a preprocessor symbol and optionally can set it to a value. This one defines the CW preprocessor symbol.

Table 3-3. Linker Options

Option	Description
-proc Zen	Generates and links object code for Zen processor. The compiler uses unsigned as the default parameter for the -char switch
-code_merging all	Removes duplicated functions to reduce object code size
-far_near_addressing	Simplifies address computations to reduce object code size and improve performance
-vle_enhance_merging	Removes duplicated functions that are called by functions that use VLE instructions to reduce object code size
-listdwarf	DWARF debugging information in the linker's map file
-sym dwarf-2,full	Generate DWARF-2-conforming debugging information (Debug With Arbitrary Record Format)
-char unsigned	Controls the default sign of the char data type: char data items are unsigned.

3.1.2 DIAB Compiler/Linker/Assembler Options

Table 3-4. Compiler Options

Option	Description
-Xdialect-ansi	Follow the ANSI C standard with some additions
-XO	Enables extra optimizations to produce highly optimized code
-g	Generate symbolic debugger information and disable some optimizations.
-Xsize-opt	Optimize for size rather than speed when there is a choice
-Xsmall-data=0	Set Size Limit for "small data" Variables to zero.
-Xaddr-sconst=0x11	Specify addressing for constant static and global variables with size less than or equal to -Xsmall-const to far-absolute.
-Xaddr-sdata=0x11	Specify addressing for non-constant static and global variables with size less than or equal to -Xsmall-data in size to far-absolute.
-Xno-common	Disable use of the "COMMON" feature so that the compiler or assembler will allocate each uninitialized public variable in the .bss section for the module defining it, and the linker will require exactly one definition of each public variable
-Xnested-interrupts	Allow nested interrupts

Table continues on the next page...

Table 3-4. Compiler Options (continued)

Option	Description
-Xdebug-dwarf2	Generate symbolic debug information in dwarf2 format
-Xdebug-local-all	Force generation of type information for all local variables
-Xdebug-local-cie	Create common information entry per module
-Xdebug-struct-all	Force generation of type information for all typedefs, struct, union and class types
-Xforce-declarations	Generates warnings if a function is used without a previous declaration
-ee1481	Generate an error when the function was used before it has been declared
-Xmacro-undefined-warn	Generates a warning when an undefined macro name occurs in a #if preprocessor directive
-W:as,-l	Pass the option "-l" (lower case letter L) to the assembler to get an assembler listing file
-Wa,-Xisa-vle	Instruct the assembler to expect and assemble VLE (Variable Length Encoding) instructions rather than BookE instructions.
-DAUTOSAR_OS_NOT_USED	-D defines a preprocessor symbol and optionally can set it to a value. AUTOSAR_OS_NOT_USED: By default in the package, the drivers are compiled to be used without Autosar OS. If the drivers are used with Autosar OS, the compiler option '-DAUTOSAR_OS_NOT_USED' must be removed from project options
-DDIAB	-D defines a preprocessor symbol and optionally can set it to a value. This one defines the DIAB preprocessor symbol.
-Xforce-prototypes	-Generate warnings if a function is used without a previous prototype declaration.
-tPPCE200Z0VEN:simple	Sets target processor to PPCE200Z0, generates ELF using EABI conventions, No floating point support (minimizes the required runtime), selects simple environment settings for Startup Module and Libraries.

Table 3-5. Assembler Options

Option	Description
-tPPCE200Z0VEN:simple	Sets target processor to PPCE200Z0, generates ELF using EABI conventions, No floating point support (minimizes the required runtime), selects simple environment settings for Startup Module and Libraries
-g	Dump the symbols in the global symbol table in each archive file.
-Xisa-vle	Expect and assemble VLE (Variable Length Encoding) instructions rather than Book E instructions. The default code section is named .text_vle instead of .text, and the default code section fill "character" is set to 0x44444444 instead of 0. The .text_vle code section will have ELF section header flags marking it as VLE code, not Book E code.

Table 3-6. Linker Options

Option	Description
-tPPCE200Z0VEN:simple	Sets target processor to PPCE200Z0, generates ELF using EABI conventions, No floating point support (minimizes the required runtime), selects simple environment settings for Startup Module and Libraries
-Xelf	Generates ELF object format for output file
-m6	Generates a detailed link map and cross reference table
-lc	Specifies to linker to search for libc.a
-Xlibc-old	Enables usage of legacy (pre-release 5.6) libraries

3.1.3 GHS Compiler/Linker/Assembler Options

Table 3-7. Compiler Options

Option	Description
-cpu=ppc560xb	Selects target processor: ppc560xb
-ansi	Specifies ANSI C with extensions. This mode extends the ANSI X3.159-1989 standard with certain useful and compatible constructs.
-noSPE	Disables the use of SPE and vector floating point instructions by the compiler.
-Ospace	Optimize for size
-sda=0	Enables the Small Data Area optimization with a threshold of 0.
-vle	Enables VLE code generation
-dual_debug	Enables the generation of DWARF, COFF, or BSD debugging information in the object file
-G	Generates source level debugging information and allows procedure call from debugger's command line.
--no_exceptions	Disables support for exception handling
-Wundef	Generates warnings for undefined symbols in preprocessor expressions
-Wimplicit-int	Issues a warning if the return type of a function is not declared before it is called
-Wshadow	Issues a warning if the declaration of a local variable shadows the declaration of a variable of the same name declared at the global scope, or at an outer scope
-Wtrigraphs	Issues a warning for any use of trigraphs
--prototype_errors	Generates errors when functions referenced or called have no prototype
--incorrect_pragma_warnings	Valid #pragma directives with wrong syntax are treated as warnings
-noslashcomment	C++ like comments will generate a compilation error
-preprocess_assembly_files	Preprocesses assembly files
--short_enum	Store enumerations in the smallest possible type
--no_commons	Allocates uninitialized global variables to a section and initializes them to zero at program startup.
-c	Produces an object file (called input-file.o) for each source file.
-DAUTOSAR_OS_NOT_USED	-D defines a preprocessor symbol and optionally can set it to a value. AUTOSAR_OS_NOT_USED: By default in the package, the drivers are compiled to be used without Autosar OS. If the drivers are used with Autosar OS, the compiler option '-DAUTOSAR_OS_NOT_USED' must be removed from project options
-DUSE_SW_VECTOR_MODE	-D defines a preprocessor symbol and optionally can set it to a value. USE_SW_VECTOR_MODE: By default in the package, drivers are compiled to be used with interrupt controller configured to be in hardware vector mode. In case of AUTOSAR_OS_NOT_USED, the compiler option "-DUSE_SW_VECTOR_MODE" must be added to the list of compiler options to be used with interrupt controller configured to be in software vector mode.
-DDISABLE_MCAL_INTERMODULE_ASR_CHECK	-D defines a preprocessor symbol to disable the inter-module version check for AR_RELEASE versions. DISABLE_MCAL_INTERMODULE_ASR_CHECK: By default in the package, drivers are compiled to perform the inter-module version check as per Autosar BSW004. When the inter-module version check needs to be disabled then the DISABLE_MCAL_INTERMODULE_ASR_CHECK global define must be added to the list of compiler options.
-DGHS	-D defines a preprocessor symbol and optionally can set it to a value. This one defines the GHS preprocessor symbol.

Table 3-8. Assembler Options

Option	Description
-cpu=ppc560xb	Selects target processor: ppc560xb

Table 3-9. Linker Options

Option	Description
-cpu=ppc560xb	Selects target processor: ppc560xb
-nostartfiles	Do not use Start files.
-vle	Enables VLE code generation
-linker_warnings	Display linker warnings

3.1.4 HT Compiler/Linker/Assembler Options

Table 3-10. Compiler Options

Option	Description
-Os	Optimize for size. '-Os' enables all '-O2' optimizations that do not typically increase code size. It also performs further optimizations designed to reduce code size.
-msdata=eabi	On System V.4 and embedded PowerPC systems, put small initialized const global and static data in the '.sdata2' section, which is pointed to by register r2. Put small initialized non-const global and static data in the '.sdata' section, which is pointed to by register r13. Put small uninitialized global and static data in the '.sbss' section, which is adjacent to the '.sdata' section.
-G1000	On embedded PowerPC systems, put global and static items less than or equal to num bytes into the small data or bss sections instead of the normal data or bss section.
-fshort-double	Use the same size for double as for float.
-mno-spe	This option used to disable spe extension (Like -noSPE on ghs compiler).
-mregnames	On System V.4 and embedded PowerPC systems do (do not) emit register names in the assembly language output using symbolic forms
-g3	Request debugging information of Level 3. Includes extra information, such as all the macro definitions present in the program.
-Wall	This enables all the warnings about constructions that some users consider questionable, and that are easy to avoid (or modify to prevent the warning), even in conjunction with macros.
-mvle	Enable support of VLE
--mcpu=z425n3	Support of e200 Z425n3 core
-DAUTOSAR_OS_NOT_USED	-D defines a preprocessor symbol and optionally can set it to a value. AUTOSAR_OS_NOT_USED: By default in the package, the drivers are compiled to be used without Autosar OS. If the drivers are used with Autosar OS, the compiler option '-DAUTOSAR_OS_NOT_USED' must be removed from project options
-DUSE_SW_VECTOR_MODE	-D defines a preprocessor symbol and optionally can set it to a value. USE_SW_VECTOR_MODE: By default in the package, drivers are compiled to be used with interrupt controller configured to be in hardware vector mode. In case of AUTOSAR_OS_NOT_USED, the compiler option "-DUSE_SW_VECTOR_MODE" must be

Table 3-10. Compiler Options

Option	Description
	added to the list of compiler options to be used with interrupt controller configured to be in software vector mode.

Table 3-11. Assembler Options

Option	Description
-c	Compile or assemble the source files, but do not link. The linking stage simply is not done. The ultimate output is in the form of an object file for each source file.
-Wa,--gdwarf2	Produce debugging information in DWARF format.
-msdata=eabi	Put small initialized const global and static data in the .sdata2 section, which is pointed to by register r2. Put small initialized non-const global and static data in the .sdata section, which is pointed to by register r13. Put small uninitialized global and static data in the .sbss section, which is adjacent to the .sdata section.
-mregnames	Do emit register names in the assembly language output using symbolic forms.
-mno-spe	This option used to disable spe extension (Like --noSPE on ghs compiler).
-mcpu=z425n3	Support of e200 z425n3 core

Table 3-12. Linker Options

Option	Description
-N	Set the text and data sections to be readable and writable. Also, do not page-align the data segment. If the output format supports Unix style magic numbers, mark the output as OMAGIC.
--no-undefined	Normally when a symbol has an undefined version, the linker will ignore it. This option disallows symbols with undefined version and a fatal error will be issued instead.
--warn-common	Normally ppc-vle-ld will give a warning if it finds an incompatible library during a library search. This option silences the warning.
--error-unresolved-symbols	This restores the linker's default behaviour of generating errors when it is reporting unresolved symbols.
-nocrt0	library crt0 from compiler library not used, it uses custom crt0 file
-nostartfiles	no start file from compiler library is used
-Wl	If the linker is being invoked indirectly, via a compiler driver (e.g. 'gcc') then all the linker command line options should be prefixed by '-Wl,'Used in combination with -no-warn-flags, --no-undefined, --warn-common and --error-unresolved-symbols
--no-warn-flags	Used in combination with -Wl disable linker warnings. See previous
-T	The -T option specifies the name of your own script ie sprig.link. The -c is a synonym of the -T option

3.2 Files required for Compilation

This section describes the include files required to compile, assemble (if assembler code) and link the MCU driver for MPC560XB microcontrollers.

To avoid integration of incompatible files, all the include files from other modules shall have the same AR_MAJOR_VERSION and AR_MINOR_VERSION, i.e. only files with the same AUTOSAR major and minor versions can be compiled.

MCU Files

- MCU_TS_T2D13M10I1R0\src\MCU.c
- MCU_TS_T2D13M10I1R0\src\MCU_Dma_LLD.c
- MCU_TS_T2D13M10I1R0\src\MCU_Ecsm_LLD.c
- MCU_TS_T2D13M10I1R0\src\MCU_eMIOS_LLD.c
- MCU_TS_T2D13M10I1R0\src\MCU_Flash_LLD.c
- MCU_TS_T2D13M10I1R0\src\MCU_Irq.c
- MCU_TS_T2D13M10I1R0\src\MCU_LLD.c
- MCU_TS_T2D13M10I1R0\src\MCU_NonASR.c
- MCU_TS_T2D13M10I1R0\src\MCU_Reg_eSys_eMIOS.c
- MCU_TS_T2D13M10I1R0\src\MCU_Siul_LLD.c
- MCU_TS_T2D13M10I1R0\src\MCU_Sscm_LLD.c
- MCU_TS_T2D13M10I1R0\include\Dma_LLD.h
- MCU_TS_T2D13M10I1R0\include\Dmamux_LLD.h
- MCU_TS_T2D13M10I1R0\include\MCU_NonASR.h
- MCU_TS_T2D13M10I1R0\include\MCU_Siul_LLD.h
- MCU_TS_T2D13M10I1R0\include\MCU_Sscm_LLD.h
- MCU_TS_T2D13M10I1R0\include\MCU_eMIOS_LLD.h
- MCU_TS_T2D13M10I1R0\include\MCU.h
- MCU_TS_T2D13M10I1R0\include\MCU_Ecsm_LLD.h
- MCU_TS_T2D13M10I1R0\include\MCU_LLD.h
- MCU_TS_T2D13M10I1R0\include\Reg_eSys_DMA.h
- MCU_TS_T2D13M10I1R0\include\Reg_eSys_DMAMUX.h
- MCU_TS_T2D13M10I1R0\include\Reg_eSys_ECSM.h
- MCU_TS_T2D13M10I1R0\include\Reg_eSys_FLASHC.h
- MCU_TS_T2D13M10I1R0\include\Reg_eSys_eMIOS.h
- MCU_TS_T2D13M10I1R0\include\Mcu_SSCM_Types.h
- MCU_TS_T2D13M10I1R0\include\Reg_eSys_SIUL.h
- MCU_TS_T2D13M10I1R0\include\Reg_eSys_SSCM.h
- MCU_TS_T2D13M10I1R0\include\Reg_eSys_STCU.h

MCU Generated Files

- MCU_TS_T2D13M10I1R0\generate_PC\src\MCU_Cfg.c (For PC Variant) - This file should be generated by the user using a configuration tool for compilation

- MCU_TS_T2D13M10I1R0\generate_PC\include\MCU_Cfg.h - This file should be generated by the user using a configuration tool for compilation
- MCU_TS_T2D13M10I1R0\generate_PC\include\Reg_eSys_MagicCarpet.h - This file should be generated by the user using a configuration tool for compilation
- MCU_TS_T2D13M10I1R0\generate_PB\src\MCU_PBcfg.c (For PB Variant) - This file should be generated by the user using a configuration tool for compilation
- MCU_TS_T2D13M10I1R0\generate\src\MCU_Dmamux_LLD.c (For PC/PB Variant) - This file should be generated by the user using a configuration tool for compilation
- MCU_TS_T2D13M10I1R0\generate\src\MCU_MC_Clocks_LLD.c (For PC/PB Variant) - This file should be generated by the user using a configuration tool for compilation
- MCU_TS_T2D13M10I1R0\generate\src\MCU_MC_Modes_LLD.c (For PC/PB Variant) - This file should be generated by the user using a configuration tool for compilation
- MCU_TS_T2D13M10I1R0\generate\src\MCU_MC_Monitor_LLD.c (For PC/PB Variant) - This file should be generated by the user using a configuration tool for compilation
- MCU_TS_T2D13M10I1R0\generate\src\MCU_MC_Power.c (For PC/PB Variant) - This file should be generated by the user using a configuration tool for compilation
- MCU_TS_T2D13M10I1R0\generate\src\MCU_MC_Reset_LLD.c (For PC/PB Variant) - This file should be generated by the user using a configuration tool for compilation

Files from Base common folder

- Base_TS_T2D13M10I1R0\include\Can_GeneralTypes.h
- Base_TS_T2D13M10I1R0\include\Cer.h
- Base_TS_T2D13M10I1R0\include\Compiler.h
- Base_TS_T2D13M10I1R0\include\Compiler_Cfg.h
- Base_TS_T2D13M10I1R0\include\ComStack_Cfg.h
- Base_TS_T2D13M10I1R0\include\ComStack_Types.h
- Base_TS_T2D13M10I1R0\include\Lin_GeneralTypes.h
- Base_TS_T2D13M10I1R0\include\Mcal.h
- Base_TS_T2D13M10I1R0\include\MemMap.h
- Base_TS_T2D13M10I1R0\include\Platform_Types.h
- Base_TS_T2D13M10I1R0\generate_PC\include\Soc_Ips.h
- Base_TS_T2D13M10I1R0\include\Std_Types.h
- Base_TS_T2D13M10I1R0\generate_PC\include\modules.h

Files from Reg folder:

- Base_TS_T2D13M10I1R0\include\Reg_eSys.h
- Base_TS_T2D13M10I1R0\include\Reg_Macros.h

Files from Dem folder:

- Dem_TS_T2D13M10I1R0\include\Dem.h
- Dem_TS_T2D13M10I1R0\include\Dem_Types.h
- Dem_TS_T2D13M10I1R0\generate_PC\include\Dem.IntErrId.h
- Dem_TS_T2D13M10I1R0\src\Dem.c

Files from Det folder:

- Det_TS_T2D13M10I1R0\include\Det.h
- Det_TS_T2D13M10I1R0\src\Det.c

Files from Rte folder:

- Rte_TS_T2D13M10I1R0\include\SchM_Adc.h
- Rte_TS_T2D13M10I1R0\src\SchM_Adc.c
- Rte_TS_T2D13M10I1R0\include\SchM_Can.h
- Rte_TS_T2D13M10I1R0\src\SchM_Can.c
- Rte_TS_T2D13M10I1R0\include\SchM_Dio.h
- Rte_TS_T2D13M10I1R0\src\SchM_Dio.c
- Rte_TS_T2D13M10I1R0\include\SchM_Fls.h
- Rte_TS_T2D13M10I1R0\src\SchM_Fls.c
- Rte_TS_T2D13M10I1R0\include\SchM_Gpt.h
- Rte_TS_T2D13M10I1R0\src\SchM_Gpt.c
- Rte_TS_T2D13M10I1R0\include\SchM_Icu.h
- Rte_TS_T2D13M10I1R0\src\SchM_Icu.c
- Rte_TS_T2D13M10I1R0\include\SchM_Mcu.h
- Rte_TS_T2D13M10I1R0\src\SchM_Mcu.c
- Rte_TS_T2D13M10I1R0\include\SchM_Port.h
- Rte_TS_T2D13M10I1R0\src\SchM_Port.c
- Rte_TS_T2D13M10I1R0\include\SchM_Pwm.h
- Rte_TS_T2D13M10I1R0\src\SchM_Pwm.c
- Rte_TS_T2D13M10I1R0\include\SchM_RamTst.h
- Rte_TS_T2D13M10I1R0\src\SchM_RamTst.c
- Rte_TS_T2D13M10I1R0\include\SchM_Spi.h
- Rte_TS_T2D13M10I1R0\src\SchM_Spi.c
- Rte_TS_T2D13M10I1R0\include\SchM_Wdg.h
- Rte_TS_T2D13M10I1R0\src\SchM_Wdg.c

3.3 Setting up the Plug-ins

The MCU driver was designed to be configured by using the EB Tresos Studio (version Elektrobit Tresos Release 14.2.1 (b140128-1223) or later).

Location of various files inside the MCU module folder:

- VSMD (Vendor Specific Module Definition) file in EB tresos Studio XDM format:
 - ..\Mcu _ TS_T2D13M10I1R0 \config\Mcu.xdm
- Pre-configuration file in EB tresos Studio XDM form containing values for reset configuration parameters:
 - ..\Mcu _ TS_T2D13M10I1R0 \config_ext\McuPreConfiguration.xdm
- VSMD (Vendor Specific Module Definition) file(s) in AUTOSAR compliant EPD format:
 - ..\Mcu _ TS_T2D13M10I1R0 \autosar\Mcu.epd
- Code Generation Templates for Pre-Compile time configuration parameters:
 - ..\Mcu _ TS_T2D13M10I1R0\generate_PC\src\MCU_Cfg.c
 - ..\Mcu _ TS_T2D13M10I1R0\generate_PC\include\MCU_Cfg.h
 - ..\Mcu _ TS_T2D13M10I1R0\generate_PC\include\Reg_eSys_MagicCarpet.h
 - ..\Mcu _ TS_T2D13M10I1R0\generate_PC\MCU_checkvalues.m
 - ..\Mcu _ TS_T2D13M10I1R0\generate_PC\MCU_RegOperations.m
- Code Generation Templates for Post-Build time configuration parameters:
 - ..\Mcu _ TS_T2D13M10I1R0\generate_PB\src\MCU_PBcfg.c
 - ..\Mcu _ TS_T2D13M10I1R0\generate_PB\MCU_checkvalues.m
 - ..\Mcu _ TS_T2D13M10I1R0\generate_PB\MCU_RegOperations.m
- Code Generation Templates for Pre-Compile/Post-Build time configuration parameters:
 - ..\Mcu _ TS_T2D13M10I1R0\generate\src\MCU_Dmamux_LLD.c
 - ..\Mcu _ TS_T2D13M10I1R0\generate\src\MCU_MC_Clocks_LLD.c
 - ..\Mcu _ TS_T2D13M10I1R0\generate\src\MCU_MC_Modes_LLD.c
 - ..\Mcu _ TS_T2D13M10I1R0\generate\src\MCU_MC_Monitor_LLD.c
 - ..\Mcu _ TS_T2D13M10I1R0\generate\src\MCU_MC_Power_LLD.c
 - ..\Mcu _ TS_T2D13M10I1R0\generate\src\MCU_MC_Reset_LLD.c

Steps to generate the configuration:

1. Copy the module folders Base_ TS_T2D13M10I1R0 , Resource_ TS_T2D13M10I1R0, Rte_ TS_T2D13M10I1R0 , Mcu_ TS_T2D13M10I1R0, Dem_ TS_T2D13M10I1R0, Det_ TS_T2D13M10I1R0 into the Tresos plugins folder.
2. Set the desired Tresos Output location folder for the generated sources and header files.
3. Use the EB tresos Studio GUI to modify ECU configuration parameters values.
4. Generate the configuration files.

For using an alternative plug-ins directory with EB tresos, please follow the steps below:

1. Copy the module folders Mcu_TS_T2D13M10I1R0 , Base_TS_T2D13M10I1R0 , Resource_TS_T2D13M10I1R0 , EcuM_TS_T2D13M10I1R0 , Mcu_TS_T2D13M10I1R0 into a directory ""<MyDirectoryPath>\<MyDirectoryName>"" under the folders ""<MyDirectoryPath>\<MyDirectoryName>\eclipse\plugins""
2. Create a text file with the extension ""<MyPluginPackageName>.link"" into the EB tresos installation in a folder called ""links"" (e.g. <MyTresosIntallation>\links \<MyPluginPackageName>.link)
3. Put the following text in that ""<MyPluginPackageName>.link"" file:
""path=<MyDirectoryPath>/<MyDirectoryName>"" , please make sure the path is described with forward slashes.
4. Use the EB tresos Studio GUI to generate the configuration files.

Chapter 4

Function calls to module

None

4.1 Function Calls during Start-up

The first BSW module to be initialized after Power on shall be MCU. The MCU shall be initialized in the following sequence.

1. Mcu_Init()
2. Mcu_InitClock()
3. Mcu_GetPllStatus () – Till PLL is locked.
4. Mcu_DistributePllClock()
5. Mcu_InitRamSection() – If required

4.2 Function Calls during Shutdown

Mcu_SetMode (sleep mode) API shall be called during GO SLEEP phase of the EcuM's Shutdown state to configure the hardware for Sleep mode. This shall be called after ICU & GPT are set to sleep.

4.3 Function Calls during Wake-up

None

Chapter 5

Module requirements

None

5.1 Exclusive areas to be defined in BSW scheduler

When `Mcu_Init_Pll0()` and `Mcu_Change_Mode()` are called, or a transition to a low power mode with flash memory powered down is requested, another write to flash registers should be prevented. In the current implementation, this is achieved through SchM API's mentioned below:

```
SchM_Enter_Mcu(SCHM_MCU_INSTANCE_0, MCU_EXCLUSIVE_AREA_00);  
SchM_Exit_Mcu(SCHM_MCU_INSTANCE_0, MCU_EXCLUSIVE_AREA_00);
```

```
SchM_Enter_Mcu(SCHM_MCU_INSTANCE_0, MCU_EXCLUSIVE_AREA_01);  
SchM_Exit_Mcu(SCHM_MCU_INSTANCE_0, MCU_EXCLUSIVE_AREA_01);
```

```
SchM_Enter_Mcu(SCHM_MCU_INSTANCE_0, MCU_EXCLUSIVE_AREA_02);  
SchM_Exit_Mcu(SCHM_MCU_INSTANCE_0, MCU_EXCLUSIVE_AREA_02);
```

```
SchM_Enter_Mcu(SCHM_MCU_INSTANCE_0, MCU_EXCLUSIVE_AREA_03);  
SchM_Exit_Mcu(SCHM_MCU_INSTANCE_0, MCU_EXCLUSIVE_AREA_03);
```

The following critical regions are used in the MCU driver:

4.1.1 MCU_EXCLUSIVE_AREA_00

Protects the write to the PFCR0, PFCR1 and PFAPR registers performed by the function `Mcu_Flash_Init` invoked by nested method calls started by the API `Mcu_Init`.

4.1.2 MCU_EXCLUSIVE_AREA_01

Protects the write to the PFCR0 and PFCR1 registers performed by the function `Mcu_Flash_Configure` invoked by nested method calls started by the API `Mcu_DistributePllClock` or `Mcu_InitClock`.

4.1.3 MCU_EXCLUSIVE_AREA_02

Protects the write to the PFCR0 and PFCR1 registers performed by the function `Mcu_Flash_Configure` invoked by nested method calls started by the API `Mcu_InitClock`.

4.1.4 MCU_EXCLUSIVE_AREA_03

Protects the transition of the flash memory to power down performed by the function `Mcu_Apply_Mode_Ram` invoked by nested method calls started by the API `Mcu_SetMode`.

Also note that FLS access to these areas should be prevented (see `FLS_EXCLUSIVE_AREA_01` - `FLS_EXCLUSIVE_AREA_04` described in the FLS IM).

5.2 Peripheral Hardware Requirements

None.

5.3 ISR to configure within OS – dependencies

The following ISR's are used by the MCU driver:

Table 5-1. MCU ISRs

ISR Name	Hardware interrupt vector
<code>Mcu_TransitionComplete_ISR</code>	52
<code>Mcu_InvalidConfig_ISR</code>	54
<code>Mcu_InvalidMode_ISR</code>	53
<code>Mcu_EnterSafeMode_ISR</code>	51
<code>RGM_AlternateMode_ISR</code>	56
<code>Mcu_XOSC_ISR</code>	57

5.4 ISR Macro

MCAL drivers use the ISR macro to define the functions that will process hardware interrupts. Depending on whether the OS is used or not, this macro can have different definitions:

a. OS is not used - AUTOSAR_OS_NOT_USED is defined:

i. If USE_SW_VECTOR_MODE is defined:

```
#define ISR(IsrName) void IsrName(void)
```

In this case, drivers' interrupt handlers are normal C functions and the prolog/epilog handle the context save and restore.

ii. If USE_SW_VECTOR_MODE is not defined:

```
#define ISR(IsrName) INTERRUPT_FUNC void IsrName(void)
```

In this case, drivers' interrupt handlers must save and restore the execution context.

Custom OS is used - AUTOSAR_OS_NOT_USED is not defined

```
#define ISR(IsrName) void OS_isr_##IsrName()
```

In this case, OS is handling the execution context when an interrupt occurs. Drivers' interrupt handlers are normal C functions.

Other vendor's OS is used - AUTOSAR_OS_NOT_USED is not defined. Please refer to the OS documentation for description of the ISR macro.

5.5 Other AUTOSAR modules - dependencies

- **Base**
- **Det:** This module is necessary for enabling Development error detection. The API function used is Det_ReportError(). The activation/deactivation of Development error detection is configurable using 'McuDevErrorDetect' configuration parameter.
- **Dem:** This module is necessary for enabling reporting of production relevant error status. The API function used is Dem_ReportErrorStatus().
- **Rte:** This module is necessary for enabling and disabling interrupts globally to implement a critical section. The API function used are SchM_Enter_Mcu() and SchM_Exit_Mcu().
- **Resource:** Sub-Derivative model is selected from Resource configuration.

Chapter 6

Memory Allocation

6.1 Sections to be defined in MemMap.h

For Post Build data:

```
#ifndef MCU_START_CONFIG_DATA_UNSPECIFIED
#define MCU_START_CONFIG_DATA_UNSPECIFIED
#define MEMMAP_ERROR
/*Memory Section for Post Build Data to be defined here. Example given in the next line*/
#pragma ghs section const=".pbMcu_cfg"
#endif
#ifndef MCU_STOP_CONFIG_DATA_UNSPECIFIED
#define MCU_STOP_CONFIG_DATA_UNSPECIFIED
#define MEMMAP_ERROR
/*End of section to be mentioned here. Example given in the next line.*/
#pragma ghs section
#endif
```

For Code:

```
#ifndef MCU_START_SEC_CODE
#define MCU_START_SEC_CODE
#define MEMMAP_ERROR
/*Memory Section for Code to be defined here.*/
#endif
#ifndef MCU_STOP_SEC_CODE
#define MCU_STOP_SEC_CODE
#define MEMMAP_ERROR
/*End of section to be mentioned here*/
#endif
#ifndef MCU_START_SEC_RAMCODE
#define MCU_START_SEC_RAMCODE
#define MEMMAP_ERROR
/*Memory Section for Code to be defined here.*/
#endif
#ifndef MCU_STOP_SEC_RAMCODE
#define MCU_STOP_SEC_RAMCODE
#define MEMMAP_ERROR
/*End of section to be mentioned here*/
#endif
```

For Variables:

```
#ifdef MCU_START_SEC_VAR_UNSPECIFIED
#undef MCU_START_SEC_VAR_UNSPECIFIED
#undef MEMMAP_ERROR
/*Memory Section for Variables to be defined here.*/
#endif
#ifdef MCU_STOP_SEC_VAR_UNSPECIFIED
#undef MCU_STOP_SEC_VAR_UNSPECIFIED
#undef MEMMAP_ERROR
/*End of section to be mentioned here*/
#endif
```

For Constant data:

```
#ifdef MCU_START_SEC_CONST_UNSPECIFIED
#undef MCU_START_SEC_CONST_UNSPECIFIED
#undef MEMMAP_ERROR
/*Memory Section for Constants to be defined here.*/
#endif
#ifdef MCU_STOP_SEC_CONST_UNSPECIFIED
#undef MCU_STOP_SEC_CONST_UNSPECIFIED
#undef MEMMAP_ERROR
/*End of section to be mentioned here*/
#endif
```

6.2 Linker command file

Memory shall be allocated for every section defined in MemMap.h

MCU_START_SEC_RAMCODE and MCU_STOP_SEC_RAMCODE are used to let the linker allocate the code found between these two defines into the RAM. They have been implemented in MemMap.h, which defines a new pragma section ".ramcode", so in the linker command file the ".ramcode" section should be defined; then a separate routine should be implemented by the user to copy this section from flash to ram, for example into startup file.

In the case of MCU driver, the functions Mcu_Flash_Init_Ram, Mcu_Flash_Configure_Ram and Mcu_Apply_Mode_Ram use these defines in order to be executed from RAM during FLASH memory configurations or transitions performed by themselves.

Chapter 7

Configuration parameters considerations

Configuration parameter class for Autosar MCU driver fall into the following variants as defined below:

7.1 Configuration Parameters

Specifies whether the configuration parameter shall be of configuration class Post Build

Table 7-1. Configuration Parameters

Configuration Container	Configuration Parameters	Configuration Variant	Current Implementation
NonAutosar	McuGetDeviceInfoApi	Pre Compile parameter for all Variants of configuration	Pre Compile
NonAutosar	McuDisableDemReportErrorStatus	Pre Compile parameter for all Variants of configuration	Pre Compile
McuGeneral	Development Error Detect	Pre Compile parameter for all Variants of configuration	Pre Compile
McuGeneral	Init clock	Pre Compile parameter for all Variants of configuration	Pre Compile
McuGeneral	Perform Reset Api	Pre Compile parameter for all Variants of Configuration	Pre Compile
McuGeneral	McuPerformResetCallout	Pre Compile parameter for all Variants of Configuration	Pre Compile
McuGeneral	Version Info Api	Pre Compile parameter for all Variants of Configuration	Pre Compile
McuGeneral	McuLPTransitionCheck	Pre Compile parameter for all Variants of Configuration	Pre Compile
McuGeneral	Mcu Get Ram State Api	Pre Compile parameter for all Variants of Configuration	Pre Compile
McuGeneral	Mcu No Pll	Pre Compile parameter for all Variants of Configuration	Pre Compile
McuGeneral	McuCalloutBeforePerformReset	Pre Compile parameter for all Variants of Configuration	Pre Compile
McuGeneral	McuConfigureFlash	Pre Compile parameter for all Variants of Configuration	Pre Compile

Table continues on the next page...

Table 7-1. Configuration Parameters (continued)

Configuration Container	Configuration Parameters	Configuration Variant	Current Implementation
McuGeneral	McuEnterLowPowerMode	Pre Compile parameter for all Variants of Configuration	Pre Compile
McuGeneral	McuSscmRAE	Pre Compile parameter for all Variants of configuration	Pre Compile
McuGeneral	McuSscmPAE	Pre Compile parameter for all Variants of configuration	Pre Compile
McuGeneral	McuConfClkOutput	Pre Compile parameter for all Variants of configuration	Pre Compile
McuGeneral	McuSkipFunction	Pre Compile parameter for all Variants of configuration	Pre Compile
McuModuleConfiguration	Operating Mode	VariantPC or VariantPB	Post Build
McuModuleConfiguration	Source Clock Failure Notification	VariantPC or VariantPB	Post Build
McuModuleConfiguration	Number Of Mcu Mode Settings	VariantPC or VariantPB	Post Build
McuModuleConfiguration	Ram Sectors	VariantPC or VariantPB	Post Build
McuModuleConfiguration	Reset Setting	VariantPC or VariantPB	Post Build
McuClockSettingConfig	External Crystal Frequency [Hz]	VariantPC or VariantPB	Post Build
McuClockSettingConfig	Loops Timeout	VariantPC or VariantPB	Post Build
McuEnableMode	STOP Mode	VariantPC or VariantPB	Post Build
McuEnableMode	STANDBY Mode	VariantPC or VariantPB	Post Build
HALT Mode	McuModeHalt	VariantPC or VariantPB	Post Build
McuEnableMode	RUN3 Mode	VariantPC or VariantPB	Post Build
McuEnableMode	RUN2 Mode	VariantPC or VariantPB	Post Build
McuEnableMode	RUN1 Mode	VariantPC or VariantPB	Post Build
McuInterruptTransition	McuTransitionComplete	VariantPC or VariantPB	Post Build
McuInterruptTransition	McuSafeMode	VariantPC or VariantPB	Post Build
McuInterruptTransition	McuInvalidMode	VariantPC or VariantPB	Post Build
McuInterruptTransition	McuInvalidConfiguration	VariantPC or VariantPB	Post Build
McuDMAMUXandEDMAConfig	eDMAChannelPriority0	VariantPC or VariantPB	Post Build
McuDMAMUXandEDMAConfig	eDMAChannelPriority4	VariantPC or VariantPB	Post Build
McuDMAMUXandEDMAConfig	eDMAChannelPriority8	VariantPC or VariantPB	Post Build
McuDMAMUXandEDMAConfig	eDMAChannelPriority12	VariantPC or VariantPB	Post Build
McuDMAMUXandEDMAConfig	ERCA	VariantPC or VariantPB	Post Build
McuDMAMUXandEDMAConfig	EDBG	VariantPC or VariantPB	Post Build
McuDMAMUXandEDMAConfig	EBW	VariantPC or VariantPB	Post Build

Table continues on the next page...

Table 7-1. Configuration Parameters (continued)

Configuration Container	Configuration Parameters	Configuration Variant	Current Implementation
McuDMAMUXandEDMAConfig	GRP0PRI	VariantPC or VariantPB	Post Build
McuDMAMUXandEDMAConfig	EMLM	VariantPC or VariantPB	Post Build
McuDMAMUXandEDMAConfig	HALT	VariantPC or VariantPB	Post Build
McuDMAMUXandEDMAConfig	HOE	VariantPC or VariantPB	Post Build
McuDMAMUXandEDMAConfig	ERGA	VariantPC or VariantPB	Post Build
McuFlashPFCR0	Read While Write Control	VariantPC or VariantPB	Post Build
McuFlashPFCR0	Port 0 Page Buffer Configuration	VariantPC or VariantPB	Post Build
McuFlashPFCR0	Port 0 Data Prefetch Enable	VariantPC or VariantPB	Post Build
McuFlashPFCR0	Port 0 Instruction Prefetch Enable	VariantPC or VariantPB	Post Build
McuFlashPFCR0	Port 0 Prefetch Limit	VariantPC or VariantPB	Post Build
McuFlashPFCR0	Port 0 Buffer Enable	VariantPC or VariantPB	Post Build
McuFlashPFCR1	Read While Write Control	VariantPC or VariantPB	Post Build
McuFlashPFCR1	Port 0 Buffer Enable	VariantPC or VariantPB	Post Build
McuFlashPFAPR	Master 0 Prefetch Disable	VariantPC or VariantPB	Post Build
McuFlashPFAPR	Master 2 Prefetch Disable	VariantPC or VariantPB	Post Build
McuFlashPFAPR	Master 0 Access Protection	VariantPC or VariantPB	Post Build
McuFlashPFAPR	Master 2 Access Protection	VariantPC or VariantPB	Post Build
McuDemEventParameterRefs	MCU E_CLOCK_FAILURE	VariantPC or VariantPB	Post Build
McuDemEventParameterRefs	MCU E_TIMEOUT_TRANSITION	VariantPC or VariantPB	Post Build
McuDemEventParameterRefs	MCU E_QUARTZ_FAILURE	VariantPC or VariantPB	Post Build
McuDemEventParameterRefs	MCU E_LOCK_FAILURE	VariantPC or VariantPB	Post Build
Mcu Clock Settings Config	McuClockSettingId	VariantPC or VariantPB	Post Build
Mcu Clock Settings Config	Mcu System Clock Frequency	VariantPC or VariantPB	Post Build
General Clock Settings	System Clock Select	VariantPC or VariantPB	Post Build
General Clock Settings	Run Mode	VariantPC or VariantPB	Post Build
General Clock Settings	XOSC	VariantPC or VariantPB	Post Build
General Clock Settings	McuPLLStatus	VariantPC or VariantPB	Post Build
General Clock Settings	Voltage Regulator	VariantPC or VariantPB	Post Build
General Clock Settings	Code Flash Status	VariantPC or VariantPB	Post Build
General Clock Settings	Data Flash Status	VariantPC or VariantPB	Post Build
General Clock Settings	RC 16Mhz	VariantPC or VariantPB	Post Build
General Clock Settings	Output power down control	VariantPC or VariantPB	Post Build
McuClockReferencePoint	ClockReferencePointFrequency	VariantPC or VariantPB	Post Build

Table continues on the next page...

Table 7-1. Configuration Parameters (continued)

Configuration Container	Configuration Parameters	Configuration Variant	Current Implementation
McuClockReferencePoint	Mcu Clock Frequency Select	VariantPC or VariantPB	Post Build
McuXtal_A	ByPass XTAL Circuits	VariantPC or VariantPB	Post Build
McuXtal_A	Counter Oscillator Value	VariantPC or VariantPB	Post Build
McuXtal_A	Xtal Mask Interrupt Status	VariantPC or VariantPB	Post Build
McuXtal_A	Xtal Divisor	VariantPC or VariantPB	Post Build
McuXtal_A	Xtal On	VariantPC or VariantPB	Post Build
McuXtal B 32KHz	ByPass XTAL Circuits	VariantPC or VariantPB	Post Build
McuXtal B 32KHz	Counter Oscillator Value	VariantPC or VariantPB	Post Build
McuXtal B 32KHz	Xtal Mask Interrupt Status	VariantPC or VariantPB	Post Build
McuXtal B 32KHz	Xtal Divisor	VariantPC or VariantPB	Post Build
McuXtal B 32KHz	Xtal On	VariantPC or VariantPB	Post Build
McuRC_16MHz	RC Divisor	VariantPC or VariantPB	Post Build
Mcu RC 128KHz	RC Divisor	VariantPC or VariantPB	Post Build
Mcu RC 128KHz	McuRCStandBy	VariantPC or VariantPB	Post Build
McuClkOutput	Clock Output	VariantPC or VariantPB	Post Build
McuClkOutput	Clock Divisor	VariantPC or VariantPB	Post Build
McuClkOutput	Clock Source	VariantPC or VariantPB	Post Build
McuPII_0	PII0 Mode	VariantPC or VariantPB	Post Build
McuPII0_Parameter	PLL0 Frequency	VariantPC or VariantPB	Post Build
McuPII0_Parameter	Output Divisor Factor	VariantPC or VariantPB	Post Build
McuPII0_Parameter	Input Divisor Factor	VariantPC or VariantPB	Post Build
McuPII0_Parameter	Multiplication Factor	VariantPC or VariantPB	Post Build
McuPII0_Parameter	Progressive Clock Switching	VariantPC or VariantPB	Post Build
McuPII0_Parameter	Fail Interrupt	VariantPC or VariantPB	Post Build
McuFmPII0_Parameter	Modulation Frequency	VariantPC or VariantPB	Post Build
McuFmPII0_Parameter	Modulation Depth	VariantPC or VariantPB	Post Build
McuFmPII0_Parameter	Spread Selection	VariantPC or VariantPB	Post Build
EMIOS Settings A	Global Prescaler	VariantPC or VariantPB	Post Build
EMIOS Settings A	Mdis Bit	VariantPC or VariantPB	Post Build
EMIOS Settings A	Global Time Base Enable	VariantPC or VariantPB	Post Build
EMIOS Settings A	Freeze Bit	VariantPC or VariantPB	Post Build
EMIOS Settings A	Global Prescaler Enable	VariantPC or VariantPB	Post Build
EMIOS Settings B	Global Prescaler	VariantPC or VariantPB	Post Build
EMIOS Settings B	Mdis Bit	VariantPC or VariantPB	Post Build
EMIOS Settings B	Global Time Base Enable	VariantPC or VariantPB	Post Build
EMIOS Settings B	Freeze Bit	VariantPC or VariantPB	Post Build
EMIOS Settings B	Global Prescaler Enable	VariantPC or VariantPB	Post Build
Mcu Peripheral Clk Set 0	Peripheral Clk Enable	VariantPC or VariantPB	Post Build
Mcu Peripheral Clk Set 0	Peripheral Divisor Factor	VariantPC or VariantPB	Post Build

Table continues on the next page...

Table 7-1. Configuration Parameters (continued)

Configuration Container	Configuration Parameters	Configuration Variant	Current Implementation
Mcu Peripheral Clk Set 1	Peripheral Clk Enable	VariantPC or VariantPB	Post Build
Mcu Peripheral Clk Set 1	Peripheral Divisor Factor	VariantPC or VariantPB	Post Build
Mcu Peripheral Clk Set 2	Peripheral Clk Enable	VariantPC or VariantPB	Post Build
Mcu Peripheral Clk Set 2	Peripheral Divisor Factor	VariantPC or VariantPB	Post Build
Mcu Clock Monitor	PII Monitor	VariantPC or VariantPB	Post Build
Mcu Clock Monitor	Mcu RCOSC Selection	VariantPC or VariantPB	Post Build
Mcu Clock Monitor	RC Divisor Factor	VariantPC or VariantPB	Post Build
Mcu Clock Monitor	HIGH Frequency Reference	VariantPC or VariantPB	Post Build
Mcu Clock Monitor	LOW Frequency Reference	VariantPC or VariantPB	Post Build
McuFlashPFCR0	Address Pipelining Control	VariantPC or VariantPB	Post Build
McuFlashPFCR0	Write Wait State Control	VariantPC or VariantPB	Post Build
McuFlashPFCR0	Read Wait State Control	VariantPC or VariantPB	Post Build
McuFlashPFCR1	Address Pipelining Control	VariantPC or VariantPB	Post Build
McuFlashPFCR1	Write Wait State Control	VariantPC or VariantPB	Post Build
McuFlashPFCR1	Read Wait State Control	VariantPC or VariantPB	Post Build
McuModeSettingConf	McuMode	VariantPC or VariantPB	Post Build
McuModeSettingConf	OperatingMode	VariantPC or VariantPB	Post Build
McuModeSettingConf	LowPowerClockSettingConfig	VariantPC or VariantPB	Post Build
McuModeSettingConf	SystemClock	VariantPC or VariantPB	Post Build
McuModeSettingConf	Mcu Voltage Regulator	VariantPC or VariantPB	Post Build
McuModeSettingConf	Mcu Boot Mode	VariantPC or VariantPB	Post Build
McuModeSettingConf	RC 16Mhz	VariantPC or VariantPB	Post Build
McuModeSettingConf	Output Power Down Control	VariantPC or VariantPB	Post Build
McuModeSettingConf	McuFastTransition	VariantPC or VariantPB	Post Build
McuModeSettingConf	McuAuxiliaryClockSettings	VariantPC or VariantPB	Post Build
GeneralClockSettings	McuOSC	VariantPC or VariantPB	Post Build
GeneralClockSettings	McuPLLStatus	VariantPC or VariantPB	Post Build
GeneralClockSettings	McuCodeFlashStatus	VariantPC or VariantPB	Post Build
GeneralClockSettings	McuDataFlashStatus	VariantPC or VariantPB	Post Build
McuRamSectorSettingConf	RamDefaultValue	VariantPC or VariantPB	Post Build
McuRamSectorSettingConf	RamSectionBaseAddress	VariantPC or VariantPB	Post Build
McuRamSectorSettingConf	RamSectionSize	VariantPC or VariantPB	Post Build
McuRamSectorSettingConf	RamSectionBaseAddrLinkerSymbol	VariantPC or VariantPB	Post Build
McuRamSectorSettingConf	RamSectionSizeLinkerSymbol	VariantPC or VariantPB	Post Build
McuDMA			
DSPIO_TX	DMA Channel Id	VariantPC or VariantPB	Post Build
DSPIO_TX	Mcu DMA Source	VariantPC or VariantPB	Post Build
DSPIO_TX	DMA Channel Enable	VariantPC or VariantPB	Post Build
DSPIO_TX	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build

Table continues on the next page...

Table 7-1. Configuration Parameters (continued)

Configuration Container	Configuration Parameters	Configuration Variant	Current Implementation
DSPI0_RX	DMA Channel Id	VariantPC or VariantPB	Post Build
DSPI0_RX	Mcu DMA Source	VariantPC or VariantPB	Post Build
DSPI0_RX	DMA Channel Enable	VariantPC or VariantPB	Post Build
DSPI0_RX	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
DSPI1_TX	DMA Channel Id	VariantPC or VariantPB	Post Build
DSPI1_TX	Mcu DMA Source	VariantPC or VariantPB	Post Build
DSPI1_TX	DMA Channel Enable	VariantPC or VariantPB	Post Build
DSPI1_TX	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
DSPI1_RX	DMA Channel Id	VariantPC or VariantPB	Post Build
DSPI1_RX	Mcu DMA Source	VariantPC or VariantPB	Post Build
DSPI1_RX	DMA Channel Enable	VariantPC or VariantPB	Post Build
DSPI1_RX	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
DSPI2_TX	DMA Channel Id	VariantPC or VariantPB	Post Build
DSPI2_TX	Mcu DMA Source	VariantPC or VariantPB	Post Build
DSPI2_TX	DMA Channel Enable	VariantPC or VariantPB	Post Build
DSPI2_TX	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
DSPI2_RX	DMA Channel Id	VariantPC or VariantPB	Post Build
DSPI2_RX	Mcu DMA Source	VariantPC or VariantPB	Post Build
DSPI2_RX	DMA Channel Enable	VariantPC or VariantPB	Post Build
DSPI2_RX	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
DSPI3_TX	DMA Channel Id	VariantPC or VariantPB	Post Build
DSPI3_TX	Mcu DMA Source	VariantPC or VariantPB	Post Build
DSPI3_TX	DMA Channel Enable	VariantPC or VariantPB	Post Build
DSPI3_TX	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
DSPI3_RX	DMA Channel Id	VariantPC or VariantPB	Post Build
DSPI3_RX	Mcu DMA Source	VariantPC or VariantPB	Post Build
DSPI3_RX	DMA Channel Enable	VariantPC or VariantPB	Post Build
DSPI3_RX	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
DSPI4_TX	DMA Channel Id	VariantPC or VariantPB	Post Build
DSPI4_TX	Mcu DMA Source	VariantPC or VariantPB	Post Build
DSPI4_TX	DMA Channel Enable	VariantPC or VariantPB	Post Build
DSPI4_TX	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
DSPI4_RX	DMA Channel Id	VariantPC or VariantPB	Post Build
DSPI4_RX	Mcu DMA Source	VariantPC or VariantPB	Post Build
DSPI4_RX	DMA Channel Enable	VariantPC or VariantPB	Post Build
DSPI4_RX	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
DSPI5_TX	DMA Channel Id	VariantPC or VariantPB	Post Build
DSPI5_TX	Mcu DMA Source	VariantPC or VariantPB	Post Build
DSPI5_TX	DMA Channel Enable	VariantPC or VariantPB	Post Build

Table continues on the next page...

Table 7-1. Configuration Parameters (continued)

Configuration Container	Configuration Parameters	Configuration Variant	Current Implementation
DSPI5_TX	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
DSPI5_RX	DMA Channel Id	VariantPC or VariantPB	Post Build
DSPI5_RX	Mcu DMA Source	VariantPC or VariantPB	Post Build
DSPI5_RX	DMA Channel Enable	VariantPC or VariantPB	Post Build
DSPI5_RX	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
EMIOS0_CH0	DMA Channel Id	VariantPC or VariantPB	Post Build
EMIOS0_CH0	Mcu DMA Source	VariantPC or VariantPB	Post Build
EMIOS0_CH0	DMA Channel Enable	VariantPC or VariantPB	Post Build
EMIOS0_CH0	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
EMIOS0_CH1	DMA Channel Id	VariantPC or VariantPB	Post Build
EMIOS0_CH1	Mcu DMA Source	VariantPC or VariantPB	Post Build
EMIOS0_CH1	DMA Channel Enable	VariantPC or VariantPB	Post Build
EMIOS0_CH1	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
EMIOS0_CH9	DMA Channel Id	VariantPC or VariantPB	Post Build
EMIOS0_CH9	Mcu DMA Source	VariantPC or VariantPB	Post Build
EMIOS0_CH9	DMA Channel Enable	VariantPC or VariantPB	Post Build
EMIOS0_CH9	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
EMIOS0_CH18	DMA Channel Id	VariantPC or VariantPB	Post Build
EMIOS0_CH18	Mcu DMA Source	VariantPC or VariantPB	Post Build
EMIOS0_CH18	DMA Channel Enable	VariantPC or VariantPB	Post Build
EMIOS0_CH18	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
EMIOS0_CH25	DMA Channel Id	VariantPC or VariantPB	Post Build
EMIOS0_CH25	Mcu DMA Source	VariantPC or VariantPB	Post Build
EMIOS0_CH25	DMA Channel Enable	VariantPC or VariantPB	Post Build
EMIOS0_CH25	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
EMIOS0_CH26	DMA Channel Id	VariantPC or VariantPB	Post Build
EMIOS0_CH26	Mcu DMA Source	VariantPC or VariantPB	Post Build
EMIOS0_CH26	DMA Channel Enable	VariantPC or VariantPB	Post Build
EMIOS0_CH26	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
EMIOS1_CH0	DMA Channel Id	VariantPC or VariantPB	Post Build
EMIOS1_CH0	Mcu DMA Source	VariantPC or VariantPB	Post Build
EMIOS1_CH0	DMA Channel Enable	VariantPC or VariantPB	Post Build
EMIOS1_CH0	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
EMIOS1_CH9	DMA Channel Id	VariantPC or VariantPB	Post Build
EMIOS1_CH9	Mcu DMA Source	VariantPC or VariantPB	Post Build
EMIOS1_CH9	DMA Channel Enable	VariantPC or VariantPB	Post Build
EMIOS1_CH9	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
EMIOS1_CH17	DMA Channel Id	VariantPC or VariantPB	Post Build
EMIOS1_CH17	Mcu DMA Source	VariantPC or VariantPB	Post Build

Table continues on the next page...

Table 7-1. Configuration Parameters (continued)

Configuration Container	Configuration Parameters	Configuration Variant	Current Implementation
EMIOS1 CH17	DMA Channel Enable	VariantPC or VariantPB	Post Build
EMIOS1 CH17	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
EMIOS1 CH18	DMA Channel Id	VariantPC or VariantPB	Post Build
EMIOS1 CH18	Mcu DMA Source	VariantPC or VariantPB	Post Build
EMIOS1 CH18	DMA Channel Enable	VariantPC or VariantPB	Post Build
EMIOS1 CH18	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
EMIOS1 CH25	DMA Channel Id	VariantPC or VariantPB	Post Build
EMIOS1 CH25	Mcu DMA Source	VariantPC or VariantPB	Post Build
EMIOS1 CH25	DMA Channel Enable	VariantPC or VariantPB	Post Build
EMIOS1 CH25	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
EMIOS1 CH26	DMA Channel Id	VariantPC or VariantPB	Post Build
EMIOS1 CH26	Mcu DMA Source	VariantPC or VariantPB	Post Build
EMIOS1 CH26	DMA Channel Enable	VariantPC or VariantPB	Post Build
EMIOS1 CH26	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
ADC0 EOC	DMA Channel Id	VariantPC or VariantPB	Post Build
ADC0 EOC	Mcu DMA Source	VariantPC or VariantPB	Post Build
ADC0 EOC	DMA Channel Enable	VariantPC or VariantPB	Post Build
ADC0 EOC	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
ADC1 EOC	DMA Channel Id	VariantPC or VariantPB	Post Build
ADC1 EOC	Mcu DMA Source	VariantPC or VariantPB	Post Build
ADC1 EOC	DMA Channel Enable	VariantPC or VariantPB	Post Build
ADC1 EOC	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
IIC RX	DMA Channel Id	VariantPC or VariantPB	Post Build
IIC RX	Mcu DMA Source	VariantPC or VariantPB	Post Build
IIC RX	DMA Channel Enable	VariantPC or VariantPB	Post Build
IIC RX	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
IIC TX	DMA Channel Id	VariantPC or VariantPB	Post Build
IIC TX	Mcu DMA Source	VariantPC or VariantPB	Post Build
IIC TX	DMA Channel Enable	VariantPC or VariantPB	Post Build
IIC TX	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
LINFLEXD0 RX	DMA Channel Id	VariantPC or VariantPB	Post Build
LINFLEXD0 RX	Mcu DMA Source	VariantPC or VariantPB	Post Build
LINFLEXD0 RX	DMA Channel Enable	VariantPC or VariantPB	Post Build
LINFLEXD0 RX	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
LINFLEXD0 TX	DMA Channel Id	VariantPC or VariantPB	Post Build
LINFLEXD0 TX	Mcu DMA Source	VariantPC or VariantPB	Post Build
LINFLEXD0 TX	DMA Channel Enable	VariantPC or VariantPB	Post Build
LINFLEXD0 TX	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
LINFLEXD1 RX	DMA Channel Id	VariantPC or VariantPB	Post Build

Table continues on the next page...

Table 7-1. Configuration Parameters (continued)

Configuration Container	Configuration Parameters	Configuration Variant	Current Implementation
LINFLEXD1 RX	Mcu DMA Source	VariantPC or VariantPB	Post Build
LINFLEXD1 RX	DMA Channel Enable	VariantPC or VariantPB	Post Build
LINFLEXD1 RX	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
LINFLEXD1 TX	DMA Channel Id	VariantPC or VariantPB	Post Build
LINFLEXD1 TX	Mcu DMA Source	VariantPC or VariantPB	Post Build
LINFLEXD1 TX	DMA Channel Enable	VariantPC or VariantPB	Post Build
LINFLEXD1 TX	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
ALWAYS ENABLED 0	DMA Channel Id	VariantPC or VariantPB	Post Build
ALWAYS ENABLED 0	Mcu DMA Source	VariantPC or VariantPB	Post Build
ALWAYS ENABLED 0	DMA Channel Enable	VariantPC or VariantPB	Post Build
ALWAYS ENABLED 0	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
ALWAYS ENABLED 1	DMA Channel Id	VariantPC or VariantPB	Post Build
ALWAYS ENABLED 1	Mcu DMA Source	VariantPC or VariantPB	Post Build
ALWAYS ENABLED 1	DMA Channel Enable	VariantPC or VariantPB	Post Build
ALWAYS ENABLED 1	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
ALWAYS ENABLED 2	DMA Channel Id	VariantPC or VariantPB	Post Build
ALWAYS ENABLED 2	Mcu DMA Source	VariantPC or VariantPB	Post Build
ALWAYS ENABLED 2	DMA Channel Enable	VariantPC or VariantPB	Post Build
ALWAYS ENABLED 2	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
ALWAYS ENABLED 3	DMA Channel Id	VariantPC or VariantPB	Post Build
ALWAYS ENABLED 3	Mcu DMA Source	VariantPC or VariantPB	Post Build
ALWAYS ENABLED 3	DMA Channel Enable	VariantPC or VariantPB	Post Build
ALWAYS ENABLED 3	DMA Channel Trigger Enable	VariantPC or VariantPB	Post Build
McuRunConfig	McuModeRun3	VariantPC or VariantPB	Post Build
McuRunConfig	McuModeRun2	VariantPC or VariantPB	Post Build
McuRunConfig	McuModeRun1	VariantPC or VariantPB	Post Build
McuRunConfig	McuModeRun0	VariantPC or VariantPB	Post Build
McuRunConfig	McuModeDRun	VariantPC or VariantPB	Post Build
McuRunConfig	McuModeSafe	VariantPC or VariantPB	Post Build
McuLowPower	McuStop	VariantPC or VariantPB	Post Build
McuLowPower	McuSTBY	VariantPC or VariantPB	Post Build
McuLowPower	McuHalt	VariantPC or VariantPB	Post Build
McuPeriphal			
DSPI0	McuPerRunConfig	VariantPC or VariantPB	Post Build
DSPI0	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
DSPI1	McuPerRunConfig	VariantPC or VariantPB	Post Build
DSPI1	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
DSPI2	McuPerRunConfig	VariantPC or VariantPB	Post Build
DSPI2	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build

Table continues on the next page...

Table 7-1. Configuration Parameters (continued)

Configuration Container	Configuration Parameters	Configuration Variant	Current Implementation
DSPI3	McuPerRunConfig	VariantPC or VariantPB	Post Build
DSPI3	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
DSPI4	McuPerRunConfig	VariantPC or VariantPB	Post Build
DSPI4	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
DSPI5	McuPerRunConfig	VariantPC or VariantPB	Post Build
DSPI5	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 8	McuPerRunConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 8	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 9	McuPerRunConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 9	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
FlexCAN0	McuPerRunConfig	VariantPC or VariantPB	Post Build
FlexCAN0	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
FlexCAN1	McuPerRunConfig	VariantPC or VariantPB	Post Build
FlexCAN1	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
FlexCAN2	McuPerRunConfig	VariantPC or VariantPB	Post Build
FlexCAN2	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
FlexCAN3	McuPerRunConfig	VariantPC or VariantPB	Post Build
FlexCAN3	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
FlexCAN4	McuPerRunConfig	VariantPC or VariantPB	Post Build
FlexCAN4	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
FlexCAN5	McuPerRunConfig	VariantPC or VariantPB	Post Build
FlexCAN5	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
DMA CH MUX	McuPerRunConfig	VariantPC or VariantPB	Post Build
DMA CH MUX	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
ADC0	McuPerRunConfig	VariantPC or VariantPB	Post Build
ADC0	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
ADC1	McuPerRunConfig	VariantPC or VariantPB	Post Build
ADC1	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
I2C0	McuPerRunConfig	VariantPC or VariantPB	Post Build
I2C0	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 0	McuPerRunConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 0	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 1	McuPerRunConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 1	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 2	McuPerRunConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 2	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 3	McuPerRunConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 3	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 4	McuPerRunConfig	VariantPC or VariantPB	Post Build

Table continues on the next page...

Table 7-1. Configuration Parameters (continued)

Configuration Container	Configuration Parameters	Configuration Variant	Current Implementation
LIN_FLEX 4	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 5	McuPerRunConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 5	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 6	McuPerRunConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 6	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 7	McuPerRunConfig	VariantPC or VariantPB	Post Build
LIN_FLEX 7	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
CTU	McuPerRunConfig	VariantPC or VariantPB	Post Build
CTU	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
CANSampler	McuPerRunConfig	VariantPC or VariantPB	Post Build
CANSampler	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
SIUL	McuPerRunConfig	VariantPC or VariantPB	Post Build
SIUL	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
WKPU	McuPerRunConfig	VariantPC or VariantPB	Post Build
WKPU	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
eMIOS 0	McuPerRunConfig	VariantPC or VariantPB	Post Build
eMIOS 0	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
eMIOS 1	McuPerRunConfig	VariantPC or VariantPB	Post Build
eMIOS 1	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
RTC API	McuPerRunConfig	VariantPC or VariantPB	Post Build
RTC API	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
PIT RTI	McuPerRunConfig	VariantPC or VariantPB	Post Build
PIT RTI	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
CMU 0	McuPerRunConfig	VariantPC or VariantPB	Post Build
CMU 0	McuPerLowPwrConfig	VariantPC or VariantPB	Post Build
McuResetSource			
JTAG	McuEnableReset	VariantPC or VariantPB	Post Build
JTAG	McuEnableInterrupt	VariantPC or VariantPB	Post Build
JTAG	McuResetPhase	VariantPC or VariantPB	Post Build
JTAG	McuResetPin	VariantPC or VariantPB	Post Build
CORE	McuEnableReset	VariantPC or VariantPB	Post Build
CORE	McuEnableInterrupt	VariantPC or VariantPB	Post Build
CORE	McuResetPhase	VariantPC or VariantPB	Post Build
CORE	McuResetPin	VariantPC or VariantPB	Post Build
SOFT	McuEnableReset	VariantPC or VariantPB	Post Build
SOFT	McuEnableInterrupt	VariantPC or VariantPB	Post Build
SOFT	McuResetPhase	VariantPC or VariantPB	Post Build
SOFT	McuResetPin	VariantPC or VariantPB	Post Build
CHKSTOP	McuEnableReset	VariantPC or VariantPB	Post Build

Table continues on the next page...

Table 7-1. Configuration Parameters (continued)

Configuration Container	Configuration Parameters	Configuration Variant	Current Implementation
CHKSTOP	McuEnableInterrupt	VariantPC or VariantPB	Post Build
CHKSTOP	McuResetPhase	VariantPC or VariantPB	Post Build
CHKSTOP	McuResetPin	VariantPC or VariantPB	Post Build
PLL	McuEnableReset	VariantPC or VariantPB	Post Build
PLL	McuEnableInterrupt	VariantPC or VariantPB	Post Build
PLL	McuResetPhase	VariantPC or VariantPB	Post Build
PLL	McuResetPin	VariantPC or VariantPB	Post Build
CMU0_OLR	McuEnableReset	VariantPC or VariantPB	Post Build
CMU0_OLR	McuEnableInterrupt	VariantPC or VariantPB	Post Build
CMU0_OLR	McuResetPhase	VariantPC or VariantPB	Post Build
CMU0_OLR	McuResetPin	VariantPC or VariantPB	Post Build
CMU0_FHL	McuEnableReset	VariantPC or VariantPB	Post Build
CMU0_FHL	McuEnableInterrupt	VariantPC or VariantPB	Post Build
CMU0_FHL	McuResetPhase	VariantPC or VariantPB	Post Build
CMU0_FHL	McuResetPin	VariantPC or VariantPB	Post Build
LVD45	McuEnableReset	VariantPC or VariantPB	Post Build
LVD45	McuEnableInterrupt	VariantPC or VariantPB	Post Build
LVD45	McuResetPhase	VariantPC or VariantPB	Post Build
LVD45	McuResetPin	VariantPC or VariantPB	Post Build
FLASH	McuEnableReset	VariantPC or VariantPB	Post Build
FLASH	McuEnableInterrupt	VariantPC or VariantPB	Post Build
FLASH	McuResetPhase	VariantPC or VariantPB	Post Build
FLASH	McuResetPin	VariantPC or VariantPB	Post Build
EXR	McuEnableReset	VariantPC or VariantPB	Post Build
EXR	McuEnableInterrupt	VariantPC or VariantPB	Post Build
EXR	McuResetPhase	VariantPC or VariantPB	Post Build
EXR	McuResetPin	VariantPC or VariantPB	Post Build
LVD12PD0	McuEnableReset	VariantPC or VariantPB	Post Build
LVD12PD0	McuEnableInterrupt	VariantPC or VariantPB	Post Build
LVD12PD0	McuResetPhase	VariantPC or VariantPB	Post Build
LVD12PD0	McuResetPin	VariantPC or VariantPB	Post Build
LVD12PD1	McuEnableReset	VariantPC or VariantPB	Post Build
LVD12PD1	McuEnableInterrupt	VariantPC or VariantPB	Post Build
LVD12PD1	McuResetPhase	VariantPC or VariantPB	Post Build
LVD12PD1	McuResetPin	VariantPC or VariantPB	Post Build
SWT	McuEnableReset	VariantPC or VariantPB	Post Build
SWT	McuEnableInterrupt	VariantPC or VariantPB	Post Build
SWT	McuResetPhase	VariantPC or VariantPB	Post Build
SWT	McuResetPin	VariantPC or VariantPB	Post Build

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Table 7-1. Configuration Parameters (continued)

Configuration Container	Configuration Parameters	Configuration Variant	Current Implementation
LVD27	McuEnableReset	VariantPC or VariantPB	Post Build
LVD27	McuEnableInterrupt	VariantPC or VariantPB	Post Build
LVD27	McuResetPhase	VariantPC or VariantPB	Post Build
LVD27	McuResetPin	VariantPC or VariantPB	Post Build
LVD27 VREG	McuEnableReset	VariantPC or VariantPB	Post Build
LVD27 VREG	McuEnableInterrupt	VariantPC or VariantPB	Post Build
LVD27 VREG	McuResetPhase	VariantPC or VariantPB	Post Build
LVD27 VREG	McuResetPin	VariantPC or VariantPB	Post Build

Chapter 8

Integration Steps

This section gives a brief overview of the steps needed for integrating MCU Driver :

- Generate the required MCU configurations. For more details refer to [Setting up the Plug-ins](#)
- Allocate proper memory sections in MemMap.h and linker command file. For more details refer to section [Sections to be defined in MemMap.h](#)
- Make sure all include files for compilation. For more details refer to section [Files required for Compilation](#)
- Map the ISRs to their vector locations. For more details refer to section [ISR to configure within OS – dependencies](#)
- Compile and build the MCU with all the dependent modules. For more details refer to section [Building the Driver](#)



Chapter 9

ISR Reference

ISR functions exported by the MCU driver.

9.1 Software specification

The following sections contains driver software specifications.

9.1.1 Define Reference

Constants supported by the MCU driver.

9.1.2 Macros Reference

Macros supported by the MCU driver.

9.1.2.1 Macro MCU_CLOCK_NOTIFICATION_CALLOUT

Prototype: `MCU_CLOCK_NOTIFICATION_CALLOUT (cmuIndex)`

9.1.3 Enum Reference

Enumeration of all constants supported by the MCU driver.

9.1.4 Function Reference

Functions supported by the MCU driver.

9.1.5 Structs Reference

Data structures supported by the MCU driver.

9.1.6 Types Reference

Types structures supported by the MCU driver.

9.1.7 Variables Reference

Variables supported by the MCU driver.

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